

- Event (E) : A subset of outcomes from the sample space (e.g. selecting a student who studies more 10 hours a week.)
- Three Event Examples from The Data set ($N=200$):
 1. A student participating in group discussion (group = discussion = 'yes').
 2. A student having an attendance rate greater than 80% (attendance > 80%).
 3. A student having an attendance rate greater
 3. A student scoring a passing grade in the final exam (final_exam pass == 'pass').

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Integrated Statistical Analysis Report

1. Understanding The Basics

→ what is probability?

• what is probability?

→ probability is a mathematical measure of the likelihood or chance that a specific event will occur. It quantifies on a scale from 0 (impossible event) to 1 (certain event).

• Key probability terminology:

• Experiment: An activity or process with an uncertain outcome (e.g. assessing a student's final exam performance).

• Sample Space (S): The set of all possible outcomes of an experiment (e.g. $S = \{\text{pass, fail}\}$ for the final exam - pass variable).

- Calculations: Randomly selecting a single student's status for group discussion assuming equal likelihood between "yes" and "No"

∴ Group Discussion = 'yes' =

$$= \frac{1}{\text{total possible outcomes}}$$

$$= \frac{1}{2} = 0.5000 \text{ (50\%)}$$

2. Types of Events

• Empirical probability Scenario:
Calculated directly from real data
based on historical observation.

• Calculation: probability of a
student passing the final exam
based on the data set:

$$P(\text{pass}) = \frac{\text{Number of students who passed}}{\text{Total number of students}} \\ = \frac{99}{200} = 0.4950 \text{ (49.5\%)}$$

• Theoretical probability Scenario

$$P(\text{pass}) = \frac{\text{Number of students who passed}}{\text{Total number of students}} \\ = \frac{99}{200}$$

• Equally Likely Scenario:

Calculated based on logical reasoning
assuming equal likelihood of all
elementary outcomes.

• Variance (σ^2):

$$\sigma^2 = \text{Var}(X) = np(1-p)$$

$$= 3 \times 0.4950 \times 0.5050$$

$$= \boxed{0.7499}$$

4. Venn Diagram in probability

Out of 200 total students in the dataset:

• Study Hours > 10 : 137 students total
(69 only + 68 both)

• Attendance $> 80\%$: 93 students total
(25 only + 68 both)

• Both Condition Met : 38 students

Contingency Table & probability Calculations

Group Discussion	Fail	Pass	total
No	47	32	79
yes	54	67	121
total	101	99	200

Probability & Calculating:

• Joint probability (Pass and yes) :

3. Random Variable & probability Distribution

Let X be a 'discrete random variable defined as the "Number of students passing the final exam out of 3 randomly selected students".

Assuming Sampling with replacement or a sufficiently large population, $X \sim \text{Binomial}(n=3, p=0.4950)$

probability Distribution Table

x (passed)	$(\binom{3}{x})(0.4950)^x(0.5050)^{3-x}$ Formula $P(X=x) =$	$P(X=x)$
0	$(\binom{3}{0})(0.4950)^0(0.5050)^3$	0.1288
1	$(\binom{3}{1})(0.4950)^1(0.5050)^2$	0.3787
2	$(\binom{3}{2})(0.4950)^2(0.5050)^1$	0.3787
3	$(\binom{3}{3})(0.4950)^3(0.5050)^0$	0.1213
Total		1.0000

Mean & Variance calculation

• Mean (μ):

$$\mu = \cancel{84} = n \cdot p = 3 \times 0.4950 \\ = 1.4850$$

$$P(\text{pairs} | 100 \text{ yes}) = 0.5537 \neq P(\text{pairs}) = 0.4250$$

Because participating in group discussion changes the probability of pairing the exam - the two events are statistically dependent.

$P(\text{pass And Group Discussion})$

$$= \frac{67}{200} = 0.3350 \text{ (33.5\%)}$$

2. Marginal probability $P(\text{pass})$:

$$P(\text{pass}) = \frac{99}{200} = 0.4950 \text{ (49.5\%)}$$

3. Condition probability $P(\text{pass} | \text{no yes})$:

$$P(\text{pass} | \text{Group Discussion}) = \frac{P(\text{pass and yes})}{P(\text{no yes})}$$

$$= \frac{67}{121} \approx 0.5537 \text{ (55.37\%)}$$

$$P(\text{pass} | \text{high}) = \frac{0.70 \times 0.80}{0.60} = \frac{0.4000}{0.60}$$

$$= \frac{4}{6} \approx 0.7778 \quad (77.78\%)$$

Conclusion : given that a student has high attendance, there is a 77.78% probability that they passed the exam.

7 7
0.480
up discussion
of passing
a statistical
test

7 Bayes' Theorem Application
give historical data:

- $P(\text{High Attendance} | \text{pass}) = 0.70$
- $P(\text{High Attendance} | \text{fail}) = 0.40$
- $P(\text{High Attendance}) = 0.60$

step-1 Derive the prior probability $P(\text{pass})$
using the Law of Total Probability:

$$P(\text{High}) = P(\text{High} | \text{pass}) \cdot P(\text{pass}) + P(\text{High} | \text{fail}) \cdot (1 - P(\text{pass}))$$

$$0.60 = 0.70 \cdot P(\text{pass}) + 0.40(1 - P(\text{pass}))$$

$$0.20 = 0.30 \cdot P(\text{pass}) \Rightarrow P(\text{pass}) = \frac{2}{3}$$

$$\approx 0.667$$

step 2: Apply Bayes' Theorem.

Find $P(\text{pass} | \text{High Attendance})$:

$$P(\text{pass} | \text{High}) = \frac{P(\text{High} | \text{pass}) \cdot P(\text{pass})}{P(\text{High})}$$

$$= \frac{0.70 \cdot \frac{2}{3}}{0.60}$$

$$= \frac{7}{9} \approx 0.778$$